NAME:

STUDENT ID:

Remember to also initial in the upper right-hand header of each page.

Due Friday Nov. 13th by the end of class.

Portfolio 9: Recursion

Note that these activities are focused on this week's theme, but they are designed to utlize skills from all your previous portfolios as well. You should use your judgement to decide which tasks are best solved with which skills you have accumulated so far.

For completing and self-correcting 70% of the below activities, you will earn 1 grade percentage point. Additionally, for participating in a *share-out of 1 of the activities, you will earn 0.5 of a grade percentage point.*

Note that some activities we will plan to start in class and others you need to start on your own. You should finish each activity before its in-class share-out in order to correct it before final submission.

Submission: To get credit for completing a prgramming activity, submit it to gradescope and answer the written questions about your algorithm, whether your program worked and what troubleshooting you can think of. You *only have to print programs you want my feedback on*. Printed programs can still be marked up by hand for syntactic or semantic corrections.

Table of Contents

This portfolio includes the following 10 activities and the report of your weekly share-out:

Part	Problem/Task	Page	In-class Start	In-class Share-out	Submission Format
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A2	Written (real world rec)	3	F 11/6	F 11/6	written
A3	Written (math rec)	5	M 11/9	M 11/9	written
A4	Written	7	oyo	M 11/9	written
A5	Written	8	oyo	M 11/9	written
A6	Proj 4 program (word sentiment)	10	oyo	F 11/13	GS, print
A7	Proj 4 program (sentiment scores)	12	oyo	F 11/13	GS, print
A8	Proj 4 program (classification)	13	oyo	F 11/13	GS, print
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A10	Proj 4 reflection	15	oyo	F 11/13	GS, print
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B2	Share-out report (optional)	17	oyo		written

A1) Prompt: What types of problems are better solved with *recursion* than *iteration*? Think of examples, do some seem harder to solve than the others?**

A2) Real-World Recursion

For each scenario below, discuss and determine:

- Base case: When does the recursion stop?
- Recursive step: How do you break the problem into a smaller version of itself?

Example for "How many socks are in a pile?" :

- Base case: If the pile is empty, return 0.
- Recursive step: Otherwise, return 1 + the number of socks in a pile that has 1 less sock.

1. How many slices are in a loaf of bread?

2. How many steps in a staircase?

3. How many people are in a line at the grocery store?

4. How many pages are in a book?

5. How many cups are in a stack?

A3) Mathematical Recursion

For each mathematical scenario below, write the base case and the recursive step
Write your asnwers as if you were defining a recursive function.

1. What is $n!$, n factorial?

2. What is the sum of integers from 1 to n?

3. What is the nth Fibonacci number?

4. What is the result of raising x to the power n, x^n?

5. What is the sum of the digits in a positive integer n?

6. What is the result of multiplying a positive integer n by another positive integer m?

A4) Written Questions

1. Evaluating expressions involving dictionaries

For each of the items given below, say whether they are an expression, a statement, or invalid. Interpret them in the context of variables `d1` and `d2` defined below.

- For expressions, also give their value
- For statements, also say what the effect of the statement is going to be
- For the invalid items, explain why they're invalid

```
d1 = {"tom":344, "anna":643, "peter":9482}
d2 = {"hans":633, "cay":123}
```

(a) `d1["anna"]` .....

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(b) `d1["hans"]` .....

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(c) `d1["cay"] = 737` .....

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(d) `d1 + d2` .....

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2. Basic dictionary manipulation

2.1 Give a statement that creates an empty dictionary and associates it with the variable name `new_dict`

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2.2 Give an expression that adds a new entry to the (already existing) dictionary some_dict. The key of that entry should be 5 and the value should be: `['pokemon', 'call of duty', 'guilty gear']`

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A5: More Written Questions

1. Examining a function**

Consider the following function definition.

```
def mystery(alist, blist):
    if len(alist) != len(blist):
        return None
A -->d = {}
    for idx in range(0, len(alist)):
B -->   d[alist[idx]] = blist[idx]
    return d
```

1.1 What happens in the line marked with `A`?

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1.2 What happens in the line marked with `B`?

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1.3 In one sentence or short paragraph: what does this function do?

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2. Dictionaries and variable values

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Consider the following program:
1.  def lists_to_dict(alist, blist):
2.     if len(alist) != len(blist):
3.        return None
4.     d = {}
5.     for idx in range(0, len(alist)):
6.        d[alist[idx]] = blist[idx]
7.     return d
8.
9.  new_dict = lists_to_dict(['a', 'b', 'c'], [100, 200, 300])
```

2.1 Draw a memory diagram that shows what we have in memory once this snippet of code has been completely executed (i.e. after line 9 has been run).

2.2 Draw a memory diagram that shows what we have in memory when line 4 has just been executed.

2.3 Draw a memory diagram that shows what we have in memory when line 6 has been executed for the first time.

A6) Project 4: Word Sentiment Dictionary

Read the entire Proj 4 assignemnt before beginning this activity.

Download the starter file `sentiment_analysis.py` and the three files with movie reviews described above from the course website.

Write a function `make_word_sentiment_dictionary`. The function should take the name of a file with movie reviews (in the format discussed above) as a parameter, and it should return a word sentiment dictionary. The keys in this dictionary should be individual words, and the values should be (small) dictionaries that assemble the following information about the word:

- the sum of the sentiment scores of the movie reviews in which the word appears (key: "total score")
- the number of movie reviews in which the word appears (key: "count")
- the average sentiment score of the movie reviews in which the word appears (key: "average score")

For example, here is what an excerpt of the word sentiment dictionary might look like:

```
{'nice': {'total score': 38, 'count': 14, 'average score': 2.7142857142857144},
'little': {'total score': 312, 'count': 154, 'average score': 2.0259740259740258},
'story': {'total score': 538, 'count': 251, 'average score': 2.143426294820717},
'process': {'total score': 41, 'count': 17, 'average score': 2.411764705882353},
...}
```

Note: convert all words to lower case before adding them to the dictionary. You can use the built-in string method `.lower()` for this purpose.

- In english (not code), what is your algorithm for your function/solution?

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- Were you able to solve this task? If not, can you think of anything else to try?

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For feedback, insert you printed program(s) here.

A7) Project 4: Predicting Sentiment Scores for Reviews

Read the entire Proj 4 assignemnt before beginning this activity.

Continuing from the previous activity… Now add a function `predict_sentiment_score`. Given a movie review as a string and a word sentiment dictionary, this function should return an estimated sentiment score for the review. It should estimate the score by averaging the scores of all the words in the review.

For example, given the review `"This movie is awesome !"` and the word sentiment dictionary created from the data in `movie_reviews_training.txt`, this function should return roughly `2.41`.

A movie review may contain some words that are not in the words sentiment dictionary that you've created. For those words, assume that they are neutral (i.e. that they have a score of 2).

- In english (not code), what is your algorithm for your function/solution?

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- Were you able to solve this task? If not, can you think of anything else to try?

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For feedback, insert you printed program(s) here.

A8) Project 4: Classifying Movie Reviews According to Their Sentiment

Read the entire Proj 4 assignemnt before beginning this activity.

Continuing from the previous activity… If you test a few of the reviews from the `movie_reviews_dev.txt` file, you will see that we are not able to totally accurately predict the sentiment scores. Most predictions your function makes are closer to the average than the actual scores of the reviews. However, if all we are interested in is whether or not a movie is rated positively, that may not matter. We may still be able to use the predicted scores to decide whether we should go see a movie or not.

Add a function `is_positive` that takes a movie review and a word sentiment dictionary, then classifies the review as either positive (i.e. humans would give it a score of 3 or 4) or not. The function should return a boolean value (i.e. `True` or `False`).

Try out your classification function on a few movie reviews from the `movie_reviews_dev.txt` file. The function won't get it right for all reviews, but it should get it right more often than not.

- In english (not code), what is your algorithm for your function/solution?

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- Were you able to solve this task? If not, can you think of anything else to try?

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For feedback, insert you printed program(s) here.

A9) Project 4: Fine Tune Your System

Read the entire Proj 4 assignemnt before beginning this activity.

Continuing from the previous activity… Now uncomment the definition of the function evaluate as well as the line in the testing area that calls it. This will run your sentiment analysis system on all 800 movie reviews from the file `movie_reviews_dev.txt` and print out some information on how many of them were classified correctly.

It should be fairly easy to classify more than 60% correctly. With further fine-tuning of `is_positive`, you may manage to get an accuracy as high as 75%.

Remember to evaluate and the line that calls it should be uncommented in your submission.

- In english (not code), what is your algorithm for your function/solution?

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- Were you able to solve this task? If not, can you think of anything else to try?

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A10) Project 4: Reflection

Compltet Proj 4 before writing this reflection.

Reflect on your work on this project and write your thoughts in a file called `reflection.pdf`. Note this reflection holds nearly half of the points for this assignment and thus should be quite substantial. Write in full sentences, in paragrpah/essay form. Include any references/citations you find as you are exploring your thoughts. Additionally, come to class ready to discuss your ideas. In particular, try to answer, based on your own opinion and in your own words:

- Can computers think?
- How do/can computers learn?
- What are some strengths and weaknesses of the method we used to simulate intelligent behavior in this project?
- How do you think the design of this program could be improved?

Type up your reflection as a short essay, print it out and insert it here.

B1: Share-out

0. Did you share-out this week? If not, and if you would like to make up your credit, where in the upcoming schedule would you like to be added?

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1. For which activity did you share-out?

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2. What was the solution you shared?

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3. If any, what changes did you make to your solution in response to questions or discussion?

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B2: Share-out (optional)

0. Did you share-out this week? If not, and if you would like to make up your credit, where in the upcoming schedule would you like to be added?

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1. For which activity did you share-out?

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2. What was the solution you shared?

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3. If any, what changes did you make to your solution in response to questions or discussion?

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